

DATA HANDLING

ASSIGNMENT – 4

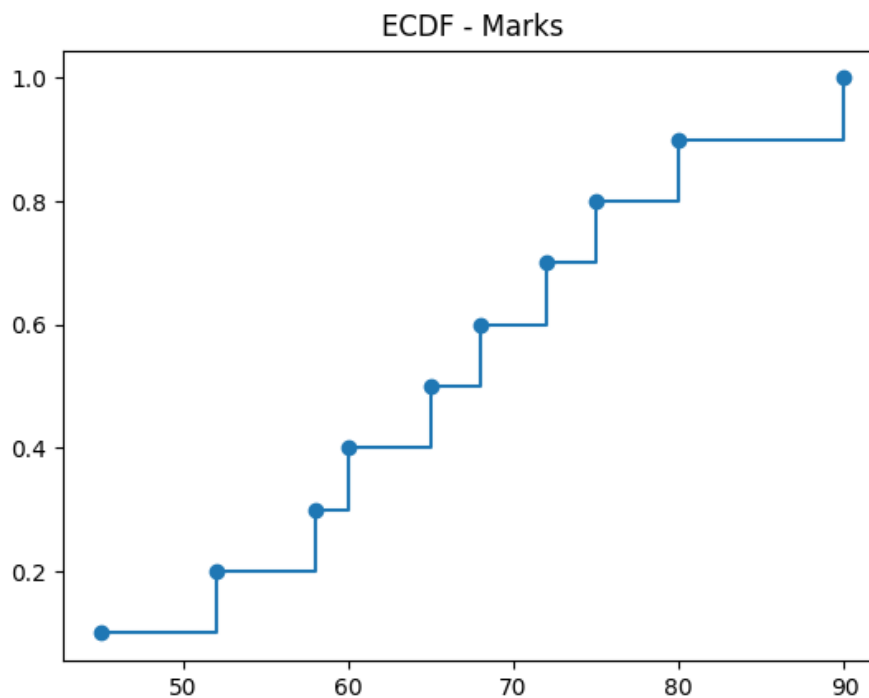
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Question 1

Question and dataset as provided by the user.

```
marks=[45,52,58,60,65,68,72,75,80,90]
import numpy as np
x=np.sort(marks)
y=np.arange(1,len(x)+1)/len(x)
plt.step(x,y,where='post'); plt.scatter(x,y)
```



Marks	ECDF
45	0.1
52	0.2

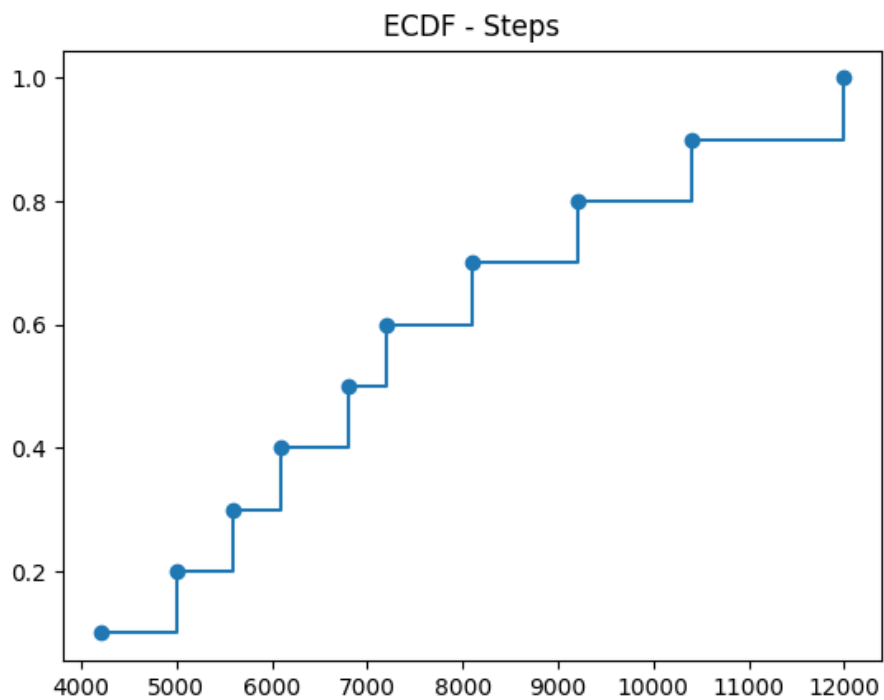
58	0.3
60	0.4
65	0.5
68	0.6
72	0.7
75	0.8
80	0.9
90	1.0

Interpretation: ECDF gives the cumulative proportion scoring at or below each mark.

Question 2

Question and dataset as provided by the user.

```
steps=[4200,...,12000]
plt.step(...)
```

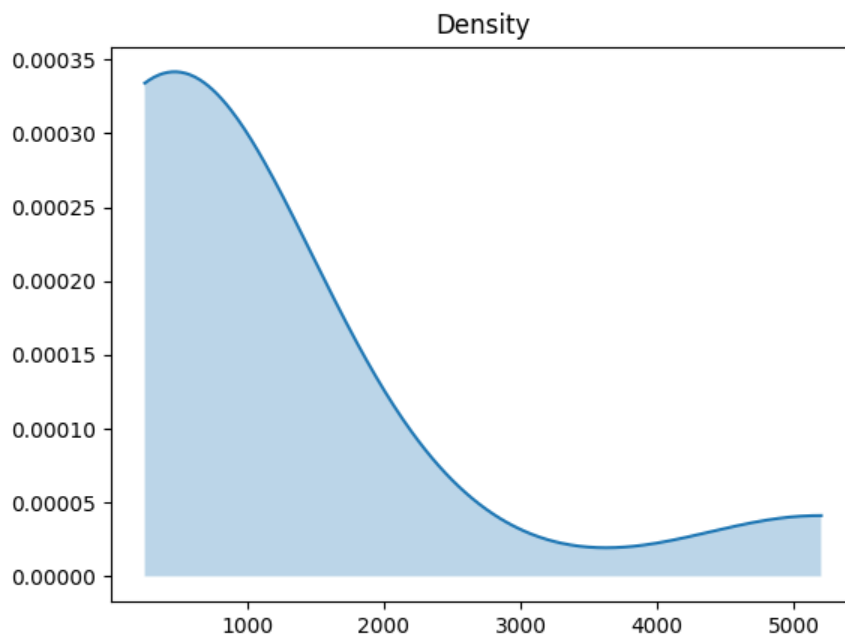
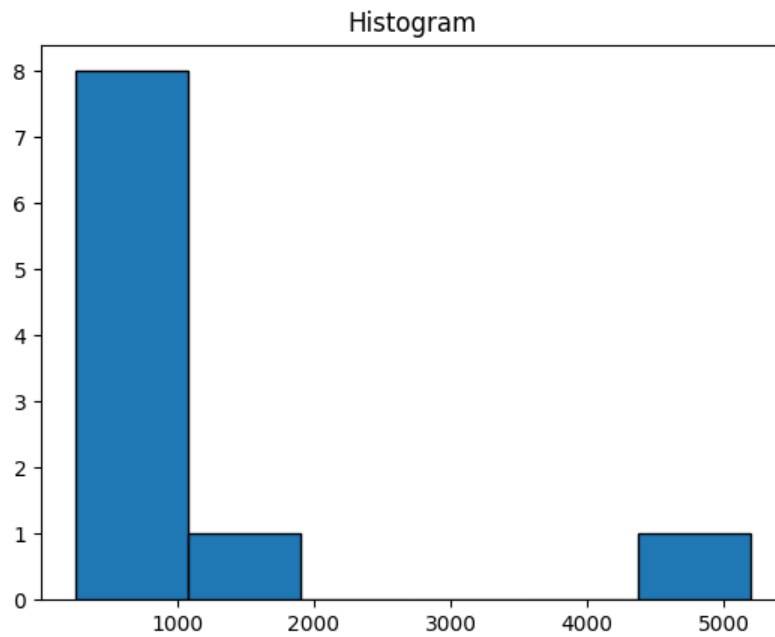


Proportion walking ≤ 8000 steps = 0.6 (70%).

Question 3

Question and dataset as provided by the user.

```
plt.hist(...)  
from scipy.stats import gaussian_kde
```

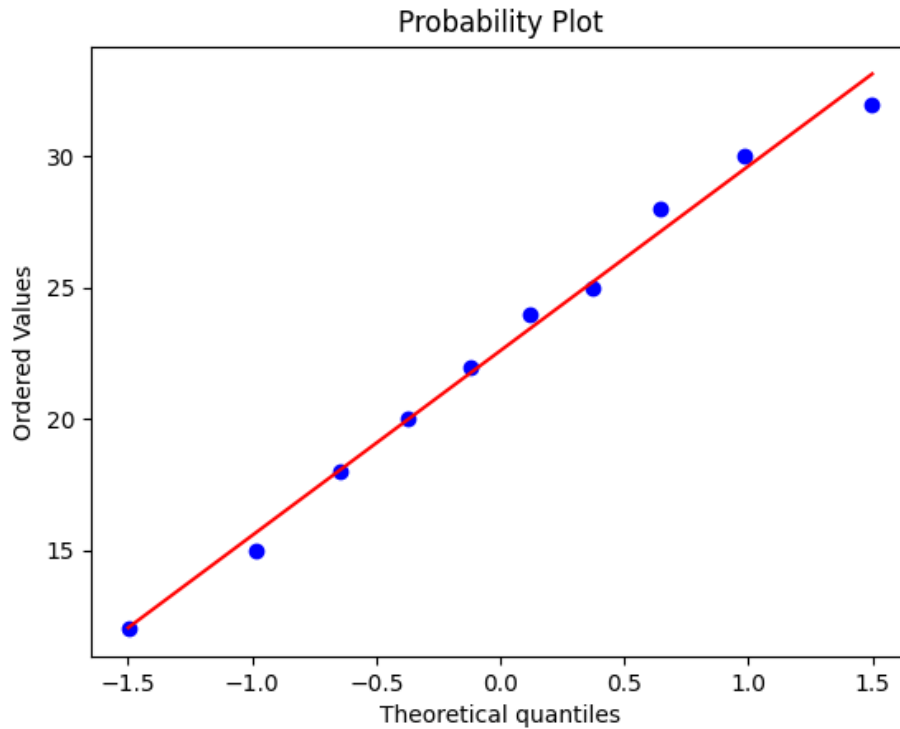


Distribution: Positively skewed due to a few extremely high purchases.

Question 4

Question and dataset as provided by the user.

```
stats.probplot(waiting_times, dist='norm', plot=plt)
```

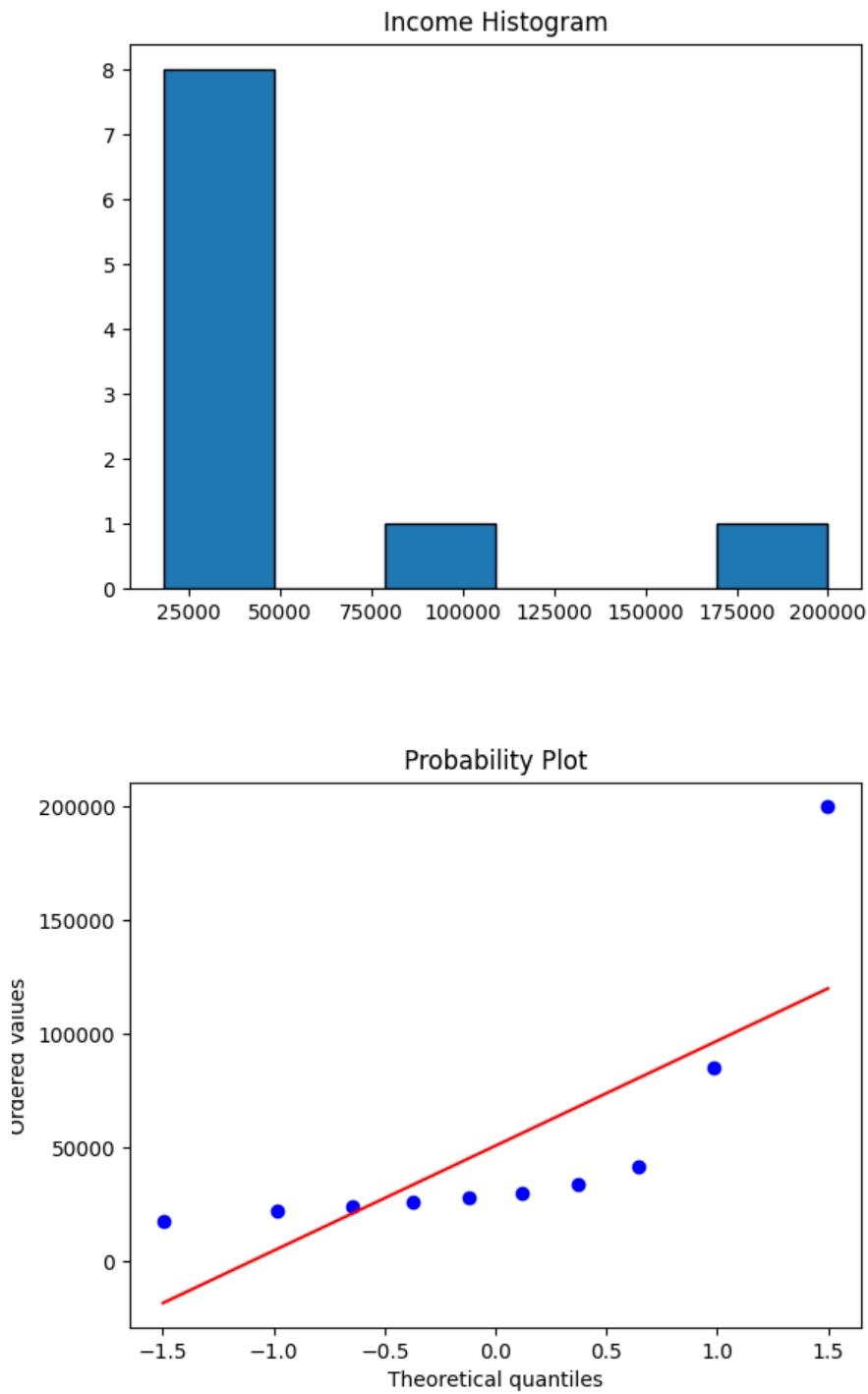


Conclusion: Points approximately follow the line, indicating approximate normality.

Question 5

Question and dataset as provided by the user.

```
plt.hist(income)
stats.probplot(income, dist='norm', plot=plt)
```



Answers:

1. Histogram plotted.
2. Q-Q plot generated.
3. Income is highly positively skewed.
4. Strong deviation from the reference line confirms non-normality.